Ritsumeikan Asia Pacific University

College of Sustainability and Tourism

Graduation Thesis

Three Margins of Electricity Recovery

Coverage, Selective Entry, and Generator Sizing in Japan's Municipal Incinerator Fleet, FY2005–FY2024

Pann Phetra

Student ID: 13223029

Supervisor: Prof. Han Ji

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Abstract

Facility counts can misdiagnose electricity recovery when activity, transition, and engineering margins are collapsed. We link 23,593 Japanese municipal- incinerator records for fiscal year (FY) 2005–FY2024 into 1,690 audited stable administrative lineages and 1,767 reported asset episodes. In FY2024, installed capacity appeared in 41.1% of all records and 46.4% of positive-throughput records; positive output appeared in 40.4% and 46.6%, respectively. Positive- output facilities handled 80.1% of throughput, while installed-capacity facilities held 70.5% of processing design capacity. First reported installed- capacity entry was sparse: 35 events entered the broad exact-year model. A frozen five-parameter Firth model gave a 300-versus-100 t/day odds ratio of 6.72 (1,999-lineage-bootstrap 95% confidence interval: 4.31–12.46). Standardized over the observed risk frame, the corresponding annual probabilities were 2.53 and 16.66 entries per 1,000 facility-years. The positive scale ordering remained stable across reported temporal-form, continuity, reporting-state, geographic- influence, and event-influence checks. Among 6,511 engineering-valid generator- years across 493 lineages, adjusted installed capacity was 79.1%, 58.6%, and 23.5% lower in pre-1990, 1990s, and 2000s cohorts than in 2010-or-later cohorts. Their electrical capacity factors were 35.3%, 22.0%, and 1.5% higher, respectively, with the 2000s interval spanning zero. The observed cohort difference therefore aligns mainly with installed generator sizing, not uniformly better annual use. The contribution is a three-margin national diagnosis; it does not identify physical projects, recoverable potential, or causal effects.

Keywords: municipal solid waste; incineration; waste-to-energy; Japan; generator sizing; capacity factor; administrative record linkage

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1 Introduction

Japan's municipal waste system relies extensively on incineration for volume reduction and hygienic treatment. Electricity recovery can use part of the heat released during combustion, but it does not by itself make incineration preferable to prevention, reuse, or recycling. Its value depends on plant design, waste properties, energy use, and the wider waste and electricity systems (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; European Commission, 2017). Japan-specific work likewise extends beyond a binary label of whether a plant generates electricity (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023).

The Ministry's FY2024 national summary reports 415 electricity-generating facilities among 991 incineration facilities, or 41.9% (Ministry of the Environment Japan, 2026). That published context is distinct from this thesis's analytical measure: 417 of 1,014 retained FY2024 facility records report positive installed electrical capacity, or 41.1%. The numerator definition and record denominator differ, so the official ratio is not substituted into the analytical calculations.

The most visible national statistic is often a facility count. Yet a small intermittent facility and a large continuous facility each contribute one unit, despite handling different waste volumes. A count describes how widely equipment appears across records, not how much waste passes through generating facilities or where processing capacity is located. Those are distinct coverage questions.

A second problem arises in facility histories. First reporting of installed capacity is uncommon and can occur within an administratively continuous facility, with a reported asset reset, or in a forward-dated record. These cases do not have one verified physical meaning. The workbooks also lack a reliable identifier over the full period: codes are absent in FY2010–FY2012 and fully change between FY2019 and FY2020. Treating them as persistent keys would break histories and create false events.

A third problem concerns gross electricity divided by throughput. This MWh/t ratio combines generator size relative to processing capacity, annual use of installed electrical capacity, and waste loading. Treating it as an independent operating measure can make a design-cohort difference look operational.

This thesis addresses the three problems in one national facility-level study. It reconstructs stable administrative lineages before creating lags, separates facility participation from waste-volume and design-capacity coverage, uses a bias-reduced event-history model for sparse first-entry events, and decomposes gross generation intensity into observable engineering components. The aim is descriptive and diagnostic. The study does not estimate the effect of a specific equipment project, infer a physical closure from administrative absence, or calculate net electricity export, useful heat, lifecycle emissions, or recoverable technical potential.

1.1 Research questions

The three research questions (RQs) are:

> **RQ1: Count versus waste-volume coverage.** How did the share of facility records reporting installed electrical-generation capacity evolve from FY2005 to FY2024, and how does that count-based measure compare with the share of recorded waste throughput handled by positive-output facilities and the share of waste-processing design capacity located at installed-capacity facilities?
>
> **RQ2: First reported installed-capacity entry.** Among stable administrative lineages first observed without installed electrical-generation capacity, which prior-year age and waste-processing-scale profiles are associated with first reporting positive capacity, and does the evidence change after the risk set is restricted to lineages with positive prior-year waste throughput?
>
> **RQ3: Installed design and annual use.** Among operating generators, how do raw installed electrical capacity and annual electrical capacity factor differ across reported start-year cohorts after observed controls, and how do these components combine with waste loading to produce gross MWh/t?

The sequence matters. RQ1 establishes the denominator problem at fleet level. RQ2 then asks which observed non-generating lineages enter the installed-capacity state. RQ3 begins only after positive throughput and gross output are observed, and asks what produces differences within the generating segment. Figure 1 shows the annual count, throughput, and design-capacity coverage measures that anchor this sequence.

1.2 Thesis objectives and significance

The thesis has four linked objectives. First, it establishes a reproducible longitudinal facility record before calculating transitions. This is necessary because the official identifiers do not provide an uninterrupted key over the study period. Second, it compares three fleet-coverage denominators rather than allowing the facility count to stand for the whole system. Third, it estimates first reported installed-capacity entry only within an explicit population at risk and uses a sparse-event estimator suited to the small number of observed entries. Fourth, it decomposes gross generation per tonne into generator design intensity, electrical capacity factor, and waste-processing utilization. The objectives therefore progress from data identity, to fleet description, to transition, and finally to conditional engineering structure.

The academic significance lies primarily in this integrated measurement architecture. Existing studies offer important analyses of system outcomes, facility performance, and

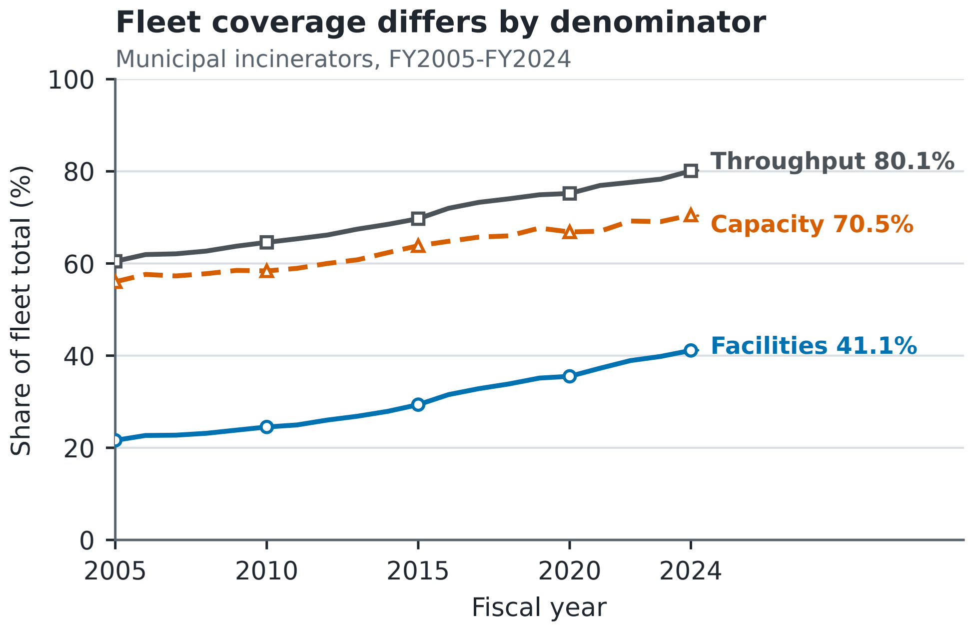

Figure 1: Facility participation, positive-output throughput coverage, and installed-capacity design-capacity coverage in Japan's municipal-incineration records, FY2005–FY2024. The denominators are intentionally different and should not be read as interchangeable shares.

Japanese waste-to-energy policy, but those questions do not share one denominator or one comparison population. Treating them as if they did can turn a descriptive fleet share into a performance statement, or a conditional generator comparison into an explanation of adoption. This thesis shows how the three questions can be studied together without collapsing their estimands. It also demonstrates why identity reconstruction is part of the research design rather than a preliminary clerical step: transition estimates depend on which annual records are considered continuous.

The practical significance is diagnostic rather than prescriptive. National and municipal decision-makers need to know whether low participation by count also means low coverage of waste activity, whether observed entry is broadly distributed or concentrated among particular facility profiles, and whether a gross-output difference reflects generator size or annual use. The results can identify where additional engineering, financial, or project-history evidence is most valuable. They cannot by themselves select a retrofit, replacement, or closure project.

The social significance follows from keeping electricity recovery within the waste hierarchy. Recovering electricity can reduce wasted combustion heat, but it does not make waste prevention, reuse, and recycling unnecessary. Clear measurement helps prevent a high throughput-coverage statistic from being presented as proof of environmental optimality. It also avoids treating every non-generating facility as an equally feasible opportunity when local waste supply, costs, grid access, heat demand, and alternative treatment arrangements remain unobserved.

2 Literature Review and Analytical Foundation

2.1 Waste-to-energy within the waste hierarchy

Waste-to-energy (WtE) is not a single environmental outcome. It is a family of thermal-treatment arrangements that can provide waste-disposal service, recover electricity or useful heat, and produce emissions and residual materials at the same time. Reviews therefore evaluate incineration within material-flow, life-cycle, energy-system, and waste-hierarchy boundaries rather than by electricity output alone (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; Lombardi et al., 2015; Münster & Meibom, 2010). The environmental value assigned to recovered energy can change with waste composition, the energy displaced, plant configuration, treatment alternatives, and the boundary chosen for analysis.

This wider framing is especially important for a study of an incineration-heavy system. Comparative policy research places prevention, reuse, and material recycling ahead of disposal, while recognizing energy recovery as one possible function of residual-waste treatment (European Commission, 2017; Sakai et al., 2011). Increasing electricity recovery is therefore not equivalent to improving the whole waste system. More generation could reflect better use of unavoidable combustion heat, but it could also coexist with high residual-waste throughput. The appropriate interpretation depends on which question is being asked.

This thesis deliberately conditions on the observed Japanese incineration fleet. It asks how electricity recovery is distributed within that fleet, not whether incineration should replace prevention or recycling. It also does not estimate life-cycle greenhouse-gas savings, avoided generation, pollutant exposure, ash management, heat use, or social welfare. Keeping these outcomes outside the estimand prevents the gross electricity measures reported below from becoming an unsupported claim of overall sustainability.

2.2 Japan's municipal transition and operating context

Japan's municipal solid-waste system combines national policy with local implementation. Municipal instruments can alter the quantity and composition of residual waste before it reaches an incinerator. For example, Japanese unit-charging programmes have been studied as waste-reduction and recycling measures, with effects depending on programme design and complementary collection arrangements (Sakai et al., 2008). This means that annual incinerator throughput is not simply an engineering input: it is partly the downstream result of local waste-management choices that are not fully observed in the facility workbooks.

Within the treatment stage, prior Japan-focused research examines several different margins. Uno (2015) describes technical developments associated with higher-efficiency power generation. Tabata and Tsai (2016) examine heat supply and the local constraints on using recovered heat. Sun et al. (2018) show through a Tokyo case that energy recovery can be considered within an integrated urban waste-management network rather than at one plant in isolation. Sasao (2018) uses repeated facility observations to study how municipal solid-waste policy relates to heat and electricity output. Shino (2019) examines generation performance relative to waste input, while Yamada et al. (2023) place the sector within longer-run Japanese decarbonization scenarios.

The institutional setting also changed during the observation window. The Ministry maintains a high-efficiency waste-power facility development manual that links facility-support requirements to generation performance and plant planning (Ministry of the Environment Japan, n.d.). The Great East Japan Earthquake and Fukushima Daiichi accident in March 2011 then changed the national energy-policy context. Japan's feed-in tariff (FIT) system began in July 2012, and waste-derived biomass electricity was among the eligible renewable categories (Agency for Natural Resources and Energy, 2017; Sasao, 2018). In FY2014, the Ministry also offered capital support for high-efficiency waste-heat recovery and waste-derived biomass generation (Ministry of the Environment Japan, 2014).

These milestones make calendar time substantively relevant, but do not create a causal design here. The workbooks do not identify subsidy receipt, FIT accreditation, electricity contracts, or investment motives. The entry model therefore includes a smooth calendar term; it does not treat FY2011 or FY2012 as an exogenous breakpoint or estimate a Fukushima or FIT effect.

Scale is important in this literature, but it is not a complete decision rule. Large, continuously operated plants can support steam conditions and generating equipment that are difficult to reproduce at small sites. Yet a heat-balance case study by Yoshida et al. (2018) identifies technically possible recovery options for a small Japanese facility and separately examines their costs and benefits. The relevant question is therefore not simply whether a small plant lacks a generator. Feasibility can also depend on waste supply, technology, capital and operating costs, grid connection, heat demand, and regional coordination.

The national administrative panel used here does not consistently observe those project-level factors. It can show which reported facility profiles are associated with witnessed entry, but it cannot identify why a municipality made a particular investment. This boundary is consequential: an association between size and entry is evidence about observed selection in the fleet, not proof that size alone caused adoption or that every large non-generator should be retrofitted.

2.3 Measuring coverage, performance, and efficiency

The literature also shows why the word *efficiency* requires qualification. Thermal-treatment reviews compare technologies using electrical, thermal, or combined energy performance and sometimes life-cycle measures (Astrup et al., 2015; Lombardi et al., 2015). The European Union's R1 formulation evaluates energy recovery using a defined accounting boundary (Grosso et al., 2010). Facility benchmarking studies may instead estimate technical or revenue efficiency relative to observed peers (Chen et al., 2012; Yeh, 2020). These are not interchangeable quantities.

The Ministry workbooks report installed electrical capacity, annual gross electricity generation, waste-processing design capacity, and annual throughput. They do not provide a complete and consistent series for net export, internal electricity use, useful heat delivery, lower heating value, steam conditions, operating cost, or emissions. Gross megawatt-hours per tonne (MWh/t) is therefore observable, but it is not a direct thermal-conversion efficiency. Shino (2019) similarly treats generation relative to waste input as informative while noting that a thermal interpretation requires calorific-value information.

Measurement also changes the apparent reach of energy recovery. A facility-count share weights a small and a large plant equally. A throughput share weights each plant by the tonnes it actually processes. A design-capacity share weights its nominal daily waste-processing capacity. None is inherently the correct denominator for every purpose. Used together, however, they reveal whether generation is diffuse across facilities or concentrated where most waste is processed and capacity is installed.

The present design consequently separates three estimands. Fleet coverage is an annual ratio whose denominator is facilities, tonnes, or waste-processing design capacity. Entry is a discrete-time conditional probability among lineages observed at risk. Generator-component estimates are conditional associations among positive-throughput, positive-output generator-years that pass stated engineering checks. A coefficient from the generator frame cannot explain why a non-generator enters, and a facility participation rate cannot describe the share of waste processed with electricity output.

This separation permits a low count share and high throughput share to be true simultaneously. It also permits newer cohorts to report higher gross MWh/t because of larger generators even when older generators do not have lower annual capacity factors. Section 3 states the exact identity used to distinguish those components.

2.4 Infrastructure heterogeneity, transition, and persistence

Incinerators are long-lived infrastructure embedded in municipal organisations, regulation, collection systems, energy networks, and local service obligations. Socio-technical research distinguishes a technology from the wider system of actors, rules, users, and material arrangements that supports it (Geels, 2004). Carbon lock-in research likewise explains how long-lived capital and institutional interdependence can make infrastructure pathways persistent (Seto et al., 2016; Unruh, 2000). These perspectives make age, cohort, and continuity relevant, but they do not establish that every old incinerator is technically or politically locked in.

This thesis uses that literature as a conceptual warning rather than a tested causal theory. The workbooks do not observe investment deliberations, sunk costs, procurement contracts, or political opposition. They also do not show whether first reported generation was installed through an in-place retrofit, a furnace replacement, a wider site redevelopment, or correction of a previous record. Accordingly, the longitudinal unit is called a stable administrative lineage, and large configuration changes create separate asset episodes. A first-entry event means the first witnessed transition from no reported generation to reported generation under a stated continuity rule; it is not automatically labelled a retrofit.

This distinction explains why identity reconstruction is substantive. If a national recoding is mistaken for facility replacement, apparent entry and exit events will be manufactured by the database. If all records sharing a site name are forced into one permanent asset, genuine replacement may be hidden. The lineage and episode sensitivity analyses translate the general infrastructure literature into a testable question: how much do transition inferences depend on the continuity definition?

2.5 What is adapted from high-profile and close comparators

High-profile studies supply research logic, not a ready-made model. Cui et al. (2026) foreground facility hierarchy, Liu et al. (2025) emphasize effectiveness over expansion, and Han et al. (2025) place recovery beside other sustainability dimensions. This thesis asks where hierarchy appears in Japan without estimating an optimization frontier, city energy-carbon system, or pollutant outcomes.

Sasao (2018) supports repeated Japanese facility observations and explicit output questions. Shino (2019) makes generation relative to waste input an important observable. This thesis retains that observable but treats it as an accounting intensity requiring decomposition.

Chen et al. (2012) and Yeh (2020) motivate separating activities within operating plants. This study adds an entry risk set and uses component regressions because the available fields do not support a comparable network or revenue frontier across all years.

Table 1: Comparator gap matrix and transparent method adaptation

Study	Unit and time structure	Primary outcome	Identity treatment	Gap retained here
Cui et al. (2026)	975 Chinese plants and 2,151 incinerators; 2023 operations and scenarios to 2035	Efficiency hierarchy and optimization	Plant/line database; entry histories not targeted	Locate hierarchy in Japan without importing an optimization frontier
Liu et al. (2025)	Chinese urban waste-energy-carbon systems	Effectiveness versus expansion	City systems, not facility linkage	Distinguish equipment diffusion from covered activity
Han et al. (2025)	Chinese configurations and recovery outcomes	Pollutant control and recovery	Configuration comparison	Keep electricity separate from unobserved pollutant and lifecycle outcomes
Sasao (2018)	Repeated Japanese incinerator observations	Policy, technology, heat, and electricity	Repeated facilities; first entry not targeted	Reconstruct lineages and an explicit non-generator risk set
Shino (2019)	22 Tokyo incinerators, FY2012–FY2017	Generation per waste input and combustion conditions	Operating-plant comparison	Decompose gross MWh/t rather than use one score
Chen et al. (2012); Yeh (2020)	Operating incineration plants	Multi-activity technical or revenue performance	Persistent plants within performance samples	Use observable components without consistent frontier inputs

The adaptation changes the unit and estimands to fit Japanese administrative data. Cui et al. supply the hierarchy logic, Sasao the closest Japan-wide repeated-facility precedent, Shino the transparent gross-generation-per-input measure, and component-performance studies the reason to separate activities. This thesis does not copy their outcome models. It reconstructs the longitudinal comparison population first and asks questions those designs leave unresolved.

2.6 Synthesis, research gap, and contribution

The reviewed literature provides strong but separated accounts of the problem. Waste-system studies define environmental and hierarchy boundaries. Japan studies document policy, technology, heat use, integrated recovery, and facility-level output. Performance studies distinguish engineering indicators from peer-relative efficiency. Transition research explains why infrastructure continuity and institutional context may matter. Close empirical comparators show the value of repeated facility observations and decomposition. Each strand answers a necessary part of the problem, but their units and outcomes differ.

The novelty claim is substantive before procedural. First, a facility count understates how much observed activity is covered. Second, first reported entry is scale-selective after sparse-event, continuity, functional-form, reporting-state, geographic, and influence checks. Third, the apparent start-cohort hierarchy in gross MWh/t is expressed mainly through installed sizing rather than uniformly better annual use. These reinterpretations change what familiar national statistics mean.

Within the literature reviewed here, the bounded integration claim is that these reinterpretations have not been connected for Japan over FY2005–FY2024 using audited lineages across coding breaks, denominator-matched coverage, an explicit first-entry risk set, and an exact engineering identity. This is not a systematic-review claim and does not assert that the individual methods are new. The contribution is the resulting diagnosis under one auditable set of estimands.

That integration matters because the three research questions can otherwise produce misleading answers. Rising facility participation does not show how much waste passes through generators. A profile associated with first entry does not explain output differences among existing generators. Gross MWh/t does not identify whether a plant has a larger generator, uses it more during the year, or processes more waste relative to that generator. The thesis therefore treats coverage, transition, and conditional performance as complementary diagnostics, not substitutes for one another and not a causal policy evaluation.

3 Data and Methods

3.1 Source workbooks and recoverable provenance

The source is the Ministry of the Environment Japan General Waste Treatment Survey facility workbook series, also distributed through the Japanese official statistics system (e-Stat, n.d.; Ministry of the Environment Japan, 2026). The study covers 20 fiscal years, FY2005–FY2024. For each year, the checked-in parser reads the first workbook sheet and searches the first six spreadsheet rows for recognized Japanese header terms. The resulting standardized fields include facility and municipality information, reported start year, waste-processing design capacity, annual throughput, furnace and facility attributes, installed electrical capacity, and gross electricity generated.

The provenance audit records all 20 workbooks, their filenames, byte sizes, 256-bit Secure Hash Algorithm (SHA-256) hashes, parsed sheet names, detected header rows, and final field mappings. The files total 15,158,836 bytes and have 20 distinct hashes. Seventeen standardized fields are detected in the older workbooks; 19 are detected from FY2018 onward. Sold-electricity and sales-revenue fields are not detected for FY2005–FY2017, which is one reason the outcome is gross generation rather than net export or revenue. The original download timestamp was not recorded. Volatile checkout modification times are deliberately not persisted; the manifest instead records each workbook's last Git commit timestamp as repository history, not retrieval time. Configured source URLs were not revalidated during the provenance build. The hashes establish which local files produced the results, not the publisher's revision history or a complete acquisition chain.

Parsing yields 23,599 raw rows. Six rows are exact duplicates of another source record and are collapsed before identity resolution, leaving 23,593 unique facility-year records. Derived age is fiscal year minus reported start year. Thirty-six source ages are missing and 355 are negative. Negative values are converted to missing for age-dependent analysis rather than set to zero. Waste-processing utilization is annual throughput divided by 365 times design capacity. Installed generation is indicated by reported electrical capacity greater than zero.

3.2 Stable administrative lineages and asset episodes

Official facility codes cannot support the longitudinal design by themselves. All 3,716 rows in FY2010–FY2012 lack a code. Between FY2019 and FY2020, both years contain codes, yet the sets have zero overlap. The latter break is a complete recode, not evidence that the entire fleet was replaced. Across that transition, the identity resolver restores 1,064 adjacent-year lineage links, equivalent to 97.3% of FY2019 records. Across the longer FY2009–FY2013 bridge, 882 official codes overlap while 1,135 stable lineages are linked.

The resolver therefore constructs a *stable administrative lineage*: a sequence of records judged to describe the same continuing administrative facility or site history. The implementation field is `stable_site_id`, but the term does not assert that ownership, buildings, furnaces, or other physical assets remain unchanged. The algorithm is deterministic and proceeds as follows.

1. Text and identifiers are normalized conservatively, including Unicode form, spacing, punctuation, and numeric-code formatting. A complete standardized row fingerprint identifies exact duplicate records, which are collapsed before matching.

2. Records are compared only within prefecture. Candidate evidence combines normalized facility name, municipality, reported start year, waste-processing capacity, furnace count, facility type, and the annual official code. Code agreement is supporting evidence, but contradictory name and configuration evidence can veto a code match.

3. Adjacent fiscal years are resolved before gaps, preventing an older history from winning a tie over an otherwise equivalent immediately prior record. Short gaps of up to four years are considered afterward. Within each prefecture-year problem, one-to-one global assignment prevents one prior record from being assigned to multiple current records.

4. Unmatched records seed new deterministic lineage IDs from canonical record fingerprints rather than row order. Within a lineage, a new asset episode is created

when reported start year changes materially in either direction, a mature record resets to a near-zero age, or a major name and configuration discontinuity appears.

The result contains 1,690 stable administrative lineages and 1,767 asset episodes, with no duplicate lineage-year and no history longer than the 20-year study window. The match audit classifies 15,324 records as code plus exact-name links, 274 as code with other supporting evidence, 6,204 as exact-name links without code support, 101 as fuzzy-name links without code support, and 1,690 as new lineage seeds.

Candidate links below the minimum score are excluded before assignment, as are ambiguous links lacking an exact-name or official-code signal. This removes 3,092 sub-threshold and 15,308 weak ambiguous candidate edges. Sixteen accepted links, spanning 14 lineages, still have a low current-record or prior-record margin but retain strong evidence; every one is exposed in the identity audit. The entry and component analyses therefore include whole-lineage identity-certain sensitivities that exclude all 14 affected lineages rather than selectively removing individual years.

Known difficult cases are embedded as executable checks. Three pairs that must link and three pairs that must remain separate are tested directly, including examples from the FY2019–FY2020 recode and earlier code reuse. The resolver is also rerun after random row permutation and after insertion of an unrelated synthetic record in six difficult or duplicate-bearing prefectures. The lineage/episode mapping must remain unchanged. These golden-link, separation, permutation, and insertion tests reduce identifiable implementation risks; they do not make probabilistic record linkage certain. Any result depending on a small number of lineages remains vulnerable to unresolved name or configuration ambiguity. The uncertainty flag makes that vulnerability testable rather than implicit.

A blinded clerical-review packet contains all 35 modeled event links, all 16 accepted uncertain links, all 31 gap links, 50 FY2019–FY2020 bridges, and a stratified accepted-pair sample (Harron et al., 2020). Scores, decisions, and final lineage IDs are withheld. Until a second reviewer completes it and disagreements are adjudicated, the packet is not independent validation.

3.3 Analytical frames and estimands

The three RQs use related but non-identical samples. Table 2 prevents their denominators from being blended.

The entry sample excludes 467 lineages already reporting positive installed capacity in their first observed year because their entry time is left-censored. A remaining lineage contributes annual risk rows until its first positive capacity observation or its last observed non-generating year. The first observed risk row is dropped from modeling because a lagged predictor is not available. The complete-covariate frame further drops 120 rows

Table 2: Analytical frames after identity and data-quality checks

Frame	Facility-year rows	Stable lineages	Events	Estimand
Full administrative panel	23,593	1,690	N/A	Annual facility, throughput, and design-capacity coverage
Descriptive non-generator risk set	16,519	1,223	55	Observed first installed-capacity entries during follow-up
Broad exact-year complete-covariate entry model	15,154	1,137	35	Conditional annual administrative-lineage entry odds using prior-year covariates
Exact-year model after prior operation	13,072	1,019	33	Nested sensitivity requiring positive prior-year throughput
Same-asset-episode continuity sensitivity	15,095	1,135	24	Entry odds after excluding transitions that cross an inferred asset-episode boundary
Identity-certain-lineage sensitivity	15,107	1,130	35	Broad model after excluding every lineage containing an accepted uncertain identity link
Two-prior-year reporting-state sensitivity	14,000	1,110	30	Requires two observed prior years with neither positive reported capacity nor output
Positive-throughput, positive-output generators	6,660	504	N/A	Descriptive operating-generator frame
Engineering-valid component model	6,511	493	N/A	Conditional generator-component associations

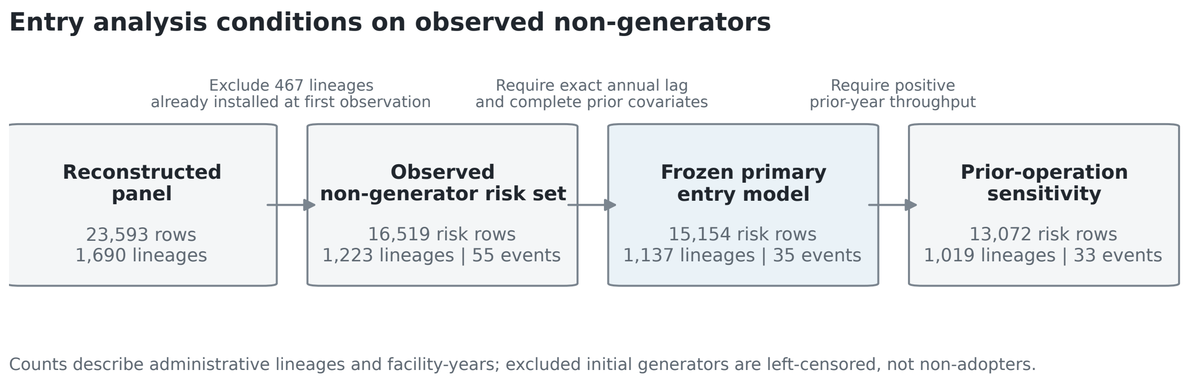

Figure 2: Entry-model sample flow. The entry analysis conditions on lineages first observed without positive installed capacity; initially installed lineages are left-censored rather than classified as non-adopters.

with missing lagged age or processing capacity, and 22 non-adjacent rows are excluded from the exact-year model; none of those 22 rows contains an event.

The 55 descriptive events, the 35 complete-covariate events, and the pathway categories answer different bookkeeping questions. Coincidentally, the pathway audit also identifies 35 continuity-lineage events. That count should not be conflated with the 35 events in the exact-year regression sample. The former is a pathway classification among observed entries; the latter is a covariate completeness restriction.

3.4 RQ1: coverage definitions and fleet identity

For fiscal year t, let N_t be all retained facility records, I_{it}^K indicate positive installed electrical capacity, I_{it}^G indicate positive gross generation with positive throughput, I_{it}^W indicate positive throughput, W_{it} be annual throughput, and C_{it} be waste-processing design capacity. To isolate denominator effects from state- definition effects, facility participation is reported as a matched two-by-two matrix:

$$P_t^{K,\text{all}} = \frac{\sum_i I_{it}^K}{N_t}, \qquad P_t^{K,\text{active}} = \frac{\sum_i I_{it}^K I_{it}^W}{\sum_i I_{it}^W}, \tag{1}$$

$$P_t^{G,\text{all}} = \frac{\sum_i I_{it}^G}{N_t}, \qquad P_t^{G,\text{active}} = \frac{\sum_i I_{it}^G}{\sum_i I_{it}^W}. \tag{2}$$

Holding the numerator state fixed makes the all-record versus active-record contrast a denominator comparison. Holding the denominator fixed shows the small difference between reported installed capacity and reported positive output. The activity-weighted coverage measures are

$$P_t^{\text{throughput}} = \frac{\sum_i W_{it} I_{it}^G}{\sum_i W_{it}}, \tag{3}$$

and

$$P_t^{\text{design}} = \frac{\sum_i C_{it} I_{it}^K}{\sum_i C_{it}}. \tag{4}$$

The throughput measure uses tonnes and the design measure tonnes per day. Positive output is used in the throughput numerator because installed capacity does not guarantee positive annual generation. For the engineering-valid subset, fleet gross intensity has the exact decomposition

$$\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}} = \left(\frac{\sum_i W_{it}^{\text{valid}}}{\sum_i W_{it}}\right) \left(\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}^{\text{valid}}}\right). \tag{5}$$

The first term is valid generator-throughput coverage and the second is throughput-weighted conditional gross intensity. This identity distinguishes how much waste is covered from how much gross electricity is reported per tonne within the covered segment.

3.5 RQ2: sparse first-entry model

The event is the first observed year with positive installed electrical capacity after a lineage has been observed without it. It is an administrative state transition. It does not uniquely identify first physical operation, equipment installation, or a particular construction history.

Positive installed capacity means a non-missing reported value greater than zero. A blank or zero field means no reported positive capacity; it is not verified physical absence. In total, 49 panel rows across 6 lineages report positive gross output with blank or zero installed capacity. None of the 35 modeled events reports positive output in its immediately prior year, and 32 report positive output in the event year. A stricter sensitivity requires two observed prior years with neither positive reported capacity nor output; it retains 14,000 rows, 1,110 lineages, and 30 events.

Discrete-time event-history analysis represents each at-risk lineage-year as a binary outcome (Allison, 1982; Beck et al., 1998). For lineage i in year t, the frozen primary model is

$$\begin{aligned}\operatorname{logit}\{\Pr(Y_{it}=1 \mid Y_{i,t-1}=0)\} = \alpha + \beta_A \frac{A_{i,t-1}}{10} + \beta_C \log\left(1+\frac{C_{i,t-1}}{100}\right) \\ + \beta_T \frac{t-2014.5}{5} + \beta_R \log(1+R_{it}).\end{aligned} \tag{6}$$

Here, A is prior-year reported age, C is prior-year processing design capacity, and R is elapsed observed time at risk. Age is scaled per ten years and calendar time per five years. Including the intercept, the model has five parameters for 35 broad-frame events and was frozen before the revised fit. The earlier 11-parameter model is retained as sensitivity. Technology and geography are not added because 35 events cannot support a larger adjustment set reliably. Their omission is a limitation: processing scale can absorb unobserved technology, regional, financial, or municipal differences. Leave-one-prefecture fits assess concentration and influence only; they are not geographic- confounding control.

Ordinary maximum-likelihood logit is vulnerable to small-sample bias and separation in a sparse event design. The primary estimator therefore uses Firth's bias reduction (Firth, 1993) and its finite-estimate solution to separation (Heinze & Schemper, 2002), maximizing the penalized log-likelihood

$$\ell_F(\theta) = \ell(\theta) + \frac{1}{2}\log|\mathcal{I}(\theta)|, \tag{7}$$

where ℓ is the binomial log-likelihood and $\mathcal{I}$ is the expected information matrix. The Jeffreys-prior penalty reduces first-order bias and keeps finite estimates under separation. The coefficient table labels the fitted-model standard errors, confidence intervals, and term-level p-values as model-based. To represent repeated observations from the same lineage, the primary uncertainty calculation uses 1,999 deterministic cluster-bootstrap replications that resample complete lineages with replacement. Percentile intervals use the resulting coefficient distributions. Every requested replication must converge and return all focal coefficients.

Four frames are fitted. The broad exact-year frame includes every complete at-risk row whose lag belongs to the same stable administrative lineage; it can cross an inferred asset-episode boundary because its estimand is first administrative-lineage entry. The prior-operation frame is a nested sensitivity that additionally requires positive prior-year throughput. It is not an independent comparison group, so differences between it and the broad frame are not treated as an equality or equivalence test. A same-asset-episode continuity sensitivity excludes the 59 exact-year rows that cross an inferred episode boundary, including 11 events. An identity-certain sensitivity excludes every lineage containing any of the 16 accepted uncertain links. These two additional frames show whether inference depends on the continuity and linkage rules.

For an intuitive scale contrast, the odds ratio comparing 300 with 100 t/day is

$$\begin{aligned} OR_{300:100} &= \exp[\beta_C \{\log(1+300/100) - \log(1+100/100)\}] \\ &= \exp(\beta_C \log 2). \end{aligned} \tag{8}$$

Because an odds ratio does not show how uncommon entry remains, standardized annual probabilities are also reported. For capacity level c, the fitted probability is averaged over the observed broad-frame distribution of age, calendar year, and elapsed risk:

$$\bar{p}(c) = \frac{1}{n} \sum_{i,t} \operatorname{logit}^{-1} \left\{ \mathbf{x}_{it}(c)' \hat{\boldsymbol{\theta}} \right\}. \tag{9}$$

This calculation sets processing capacity to 100 or 300 t/day for every risk row while retaining its other observed predictors. Percentile intervals repeat the standardization for all 1,999 lineage-bootstrap coefficient draws. These are adjusted descriptive probabilities for the observed risk population, not predicted effects of physically enlarging a facility.

The earlier 11-parameter Firth model and conventional links remain specification sensitivities.

Three additional diagnostics retain the same five-parameter structure. The capacity term is replaced by either $\log(1+C)$ or linear $C/100$ and translated to its own 300-versus-100 t/day contrast. The model is also refitted after omitting each of the 21 prefectures containing an event, and on the two-prior-year reporting-state frame. These diagnostics use model-based intervals; they test functional-form, geographic, and state-definition fragility but do not replace the 1,999-lineage-bootstrap primary inference.

3.6 RQ3: engineering decomposition and component models

Let G_{it} be annual gross generation in MWh, W_{it} annual throughput in tonnes, K_{it} installed electrical capacity in kW, and C_{it} waste-processing design capacity in t/day. Define

$$Y_{it} = \frac{G_{it}}{W_{it}} \quad \text{(gross generation intensity, MWh/t)}, \tag{10}$$

$$D_{it} = \frac{K_{it}}{C_{it}} \quad \text{(generator design intensity, kW per t/day)}, \tag{11}$$

$$F_{it} = \frac{G_{it}}{8.76K_{it}} \quad \text{(annual electrical capacity factor)}, \tag{12}$$

$$U_{it} = \frac{W_{it}}{365C_{it}} \quad \text{(waste-processing utilization)}. \tag{13}$$

The factor 8.76 converts one kW operating for 8,760 hours into annual MWh. These definitions imply the exact facility-year identity

$$Y_{it} = \frac{8.76}{365}\frac{D_{it}F_{it}}{U_{it}} = 0.024\frac{D_{it}F_{it}}{U_{it}}. \tag{14}$$

Gross MWh/t is thus a ratio produced jointly by installed sizing, annual use of that electrical capacity, and annual waste loading. The identity does not say that any one component is exogenous. For example, throughput, capacity factor, maintenance, and waste composition may be jointly determined during a year.

The primary sample requires positive installed capacity, throughput, and gross output. Values are excluded rather than clipped if gross intensity falls outside 0.01–0.80 MWh/t, electrical capacity factor outside 0.02–1.20, waste-processing utilization outside 0.02–1.20, or generator design intensity outside 0.1–100 kW per t/day. A non-missing non-negative reported age and all model fields are also required. Of 6,660 positive-throughput, positive-output rows, 149 fail at least one check, leaving 6,511 rows across 493 lineages. Heating value between 3 and 25 megajoules per kilogram (MJ/kg) is audited as a plausibility field but is not an inclusion condition for the primary component models; 102 otherwise valid rows lack it.

Two pooled component models are estimated by ordinary least squares with standard errors clustered by stable lineage (Wooldridge, 2010). Installed electrical capacity is modeled in its raw reported unit:

$$\log K_{it} = \alpha_K + V_{it}'\beta_K + \beta_{KC}\log C_{it} + T_{it}'\eta_K + \lambda_t + \varepsilon_{Kit}, \tag{15}$$

$$\begin{aligned}\log F_{it} = \alpha_F + V_{it}'\beta_F + \beta_{FC}\log C_{it} + \beta_U U_{it} \\ + T_{it}'\eta_F + \lambda_t + \varepsilon_{Fit}.\end{aligned} \tag{16}$$

V_{it} contains reported start-year cohorts before 1990, 1990–1999, and 2000–2009 relative to 2010 or later. Reported start year is an administrative design-vintage marker, not a verified boiler or generator installation date. T includes furnace count and coarse furnace and facility-type groups; λ_t are fiscal-year indicators. Because $\log D = \log K - \log C$, the design-intensity scale elasticity is $\beta_{KC} - 1$ under identical controls; this is algebraic, not independent corroboration.

A specification diagnostic compares a legacy-style regression of $\log Y$ on reported age, processing scale, waste utilization, heating value, technology, and year with the same model after adding $\log D$. Both specifications use the 5,806 engineering-valid rows with reported heating value in the 3–25 MJ/kg plausibility range. This is not a causal mediation design. It asks whether coefficients previously attributed to age, scale, or utilization are stable after the omitted installed-sizing component is represented.

Robustness checks split the period into FY2005–FY2014 and FY2015–FY2024, give each lineage equal total weight, and apply conservative and broad predefined engineering bounds. Asset-episode fixed effects and exact-adjacent first differences are used only for operating components that vary meaningfully within an episode. Design intensity is predominantly a between-asset attribute, so within-episode change is not presented as its primary estimand.

Finally, event pathways are classified from observed administrative continuity. A continuity-lineage entry remains in the same lineage and asset episode across adjacent years without a reported reset. A rebuild/replacement-like entry has an asset-episode, reported-start-year, or mature-to-new age reset. A forward-dated/placeholder entry has a future start year or new-build placeholder name. These labels are descriptive evidence rules. They do not verify the physical project mechanism. First-complete-year outcomes are compared as within-fiscal-year percentile ranks among engineering-valid generators, which reduces confounding by fleet-wide time trends but does not remove selection into each pathway.

4 Results

4.1 RQ1: facility counts understate waste-volume coverage

The official 415/991 ratio (41.9%) describes the Ministry's published count of electricity-generating facilities. The results below use the separate analytical definition of positive installed capacity among retained facility records.

Installed-capacity participation rises steadily from 21.6% of facility records in FY2005 to 41.1% in FY2024. Positive-output facilities already handle a much larger share of waste: throughput coverage rises from 60.5% to 80.1% over the same period. The design-capacity share at installed-capacity facilities moves from 56.0% to 70.5%. Figure 1 therefore shows long-run increases with year-to-year variation under all three denominators, alongside a persistent count-volume gap.

The FY2024 cross-section makes the distinction concrete. The administrative panel contains 1,014 facility records, of which 417 report positive installed electrical capacity, giving the 41.1% all-record participation rate. Among the 879 records with positive throughput, 408 report installed capacity, or 46.4%. Holding the installed-capacity state fixed therefore isolates a 5.3-percentage- point denominator difference. Positive output appears in 410 records: 40.4% of all records and 46.6% of positive-throughput records. The two output percentages use the same numerator state and isolate a 6.2-point denominator difference.

The 410 positive-output facilities process 24.70 million of the 30.84 million recorded tonnes, or 80.1%. In contrast, 469 operating non-generators process 6.14 million tonnes, or 19.9%. Another 126 records report neither positive throughput nor generation, and nine installed-capacity records report no positive output. Two FY2024 records report positive output without positive installed capacity. Installed-capacity and positive-output states are therefore reported separately rather than forced into one binary partition.

After the predefined output and capacity-factor checks, valid generator rows cover 79.7% of FY2024 throughput. Their throughput-weighted conditional gross intensity is 0.425 MWh/t. Multiplying 0.797 by 0.425 gives 0.338 MWh per total fleet tonne, exactly matching valid gross generation divided by all recorded throughput. This arithmetic is more informative than the facility count alone: most recorded waste is already processed at facilities reporting positive generation, even though installed equipment appears in fewer than half of facility records.

The result does not imply that the remaining 19.9% of operating throughput can or should all be redirected or equipped. It also does not measure electricity used internally, net export, heat recovery, marginal emissions, or project cost. Its contribution is denominator discipline: installed-capacity participation is 41.1% among all records and 46.4% among active records, while positive-output participation is 40.4% and 46.6%, respectively. Positive- output facilities cover 80.1% of tonnes, and installed-capacity facilities cover 70.5% of design capacity. Matching on activity narrows the count-volume contrast but does not remove it.

4.2 RQ2: entry is rare and strongly associated with processing scale

The descriptive risk set contains 55 first reported installed-capacity entries. The 35 exact-year modeled events comprise 24 continuity-lineage and 11 rebuild/replacement-like entries; none is forward-dated. The four calendar eras contain 7, 4, 14, and 10 modeled events, respectively.

Only 35 events remain after requiring an exact one-year lag and complete prior age and capacity; 33 remain after requiring positive prior-year throughput. This sparse count is central to interpretation. The Firth model addresses finite-estimate and first-order bias problems, but it cannot create information that the panel does not contain.

Observed rates already show strong scale ordering. Entry occurs in 1 of 3,854 risk rows in the smallest processing-capacity quartile (0.026%), 2 of 4,175 in the second quartile (0.048%), 9 of 3,702 in the third (0.243%), and 23 of 3,423 in the largest (0.672%). Because risk duration, calendar period, and age differ across these groups, the adjusted model is needed before interpreting the gradient.

In the frozen five-parameter broad Firth model, the coefficient on $\log(1 + C/100)$ is 2.749. The implied 300-versus-100 t/day odds ratio is 6.72 (1,999-lineage-bootstrap 95% CI 4.31–12.46). The prior-operation, same-episode, and identity-certain odds ratios are 7.09 (4.08–13.76), 7.15 (4.44–14.05), and 6.76 (4.23–12.30), respectively.

The adjusted absolute scale is much smaller than the odds ratio alone may suggest. Standardizing over all 15,154 broad-frame risk rows gives 2.53 annual entries per 1,000 facility-years at 100 t/day (bootstrap 95% CI 1.73–3.52) and 16.66 at 300 t/day (9.51–29.66). The adjusted difference is 14.13 entries per 1,000 facility-years (7.38–26.77). Entry therefore remains uncommon at both capacities even though its conditional odds are strongly ordered by scale.

The broad age coefficient is -0.327 per ten years (bootstrap CI -0.774 to 0.070); prior-operation and identity-certain intervals also span zero. The same-episode estimate is -0.751 (CI -1.364 to -0.206) but uses only 24 events. Age is therefore continuity-sensitive. Reclassifying each event leaves scale odds ratios from 6.12 to 7.30; deleting each event's entire lineage leaves 6.13 to 7.30. No one event or event lineage creates the scale result.

Alternative capacity forms also remain positive. Using $\log(1+C)$ gives an OR of 5.01 (model-based 95% CI 2.85–8.83), while linear capacity gives 4.22 (2.69–6.61). Omitting each event prefecture gives 6.14–7.18 across 21 fits. The stricter 30-event reporting-state frame gives 6.21 (3.26–11.81). These are diagnostics with model-based intervals; only the frozen primary uses the 1,999-lineage bootstrap for headline uncertainty.

Table 3: Consolidated entry specification audit

Specification	Events or fits	300-versus-100 t/day OR	Purpose and inference
Frozen five-parameter broad frame	35	6.72 (4.31–12.46)	Primary; 1,999-lineage bootstrap
Prior-operation frame	33	7.09 (4.08–13.76)	Positive prior-year throughput
Same-episode frame	24	7.15 (4.44–14.05)	Episode-boundary continuity
Identity-certain frame	35	6.76 (4.23–12.30)	Uncertain-link exclusion
Flexible era/duration Firth model	35	6.13 (3.92–11.21)	11-parameter temporal-form check; 499 whole-lineage bootstrap replications
Alternative capacity forms	2 fits	4.22–5.01	Functional form; model-based lower CI bounds 2.69–2.85
Two-prior-year reporting state	30	6.21 (3.26–11.81)	Stricter prior-state evidence
Leave-one-event-prefecture fits	21 fits	6.14–7.18	Geographic influence, not confounding control
Event or event-lineage attacks	70 fits	6.12–7.30	Single-event influence

Table 3 distinguishes bootstrap headline inference from model-based and deletion diagnostics. A stable direction across these rows does not recover unobserved technology, finance, policy exposure, or project history omitted from the model.

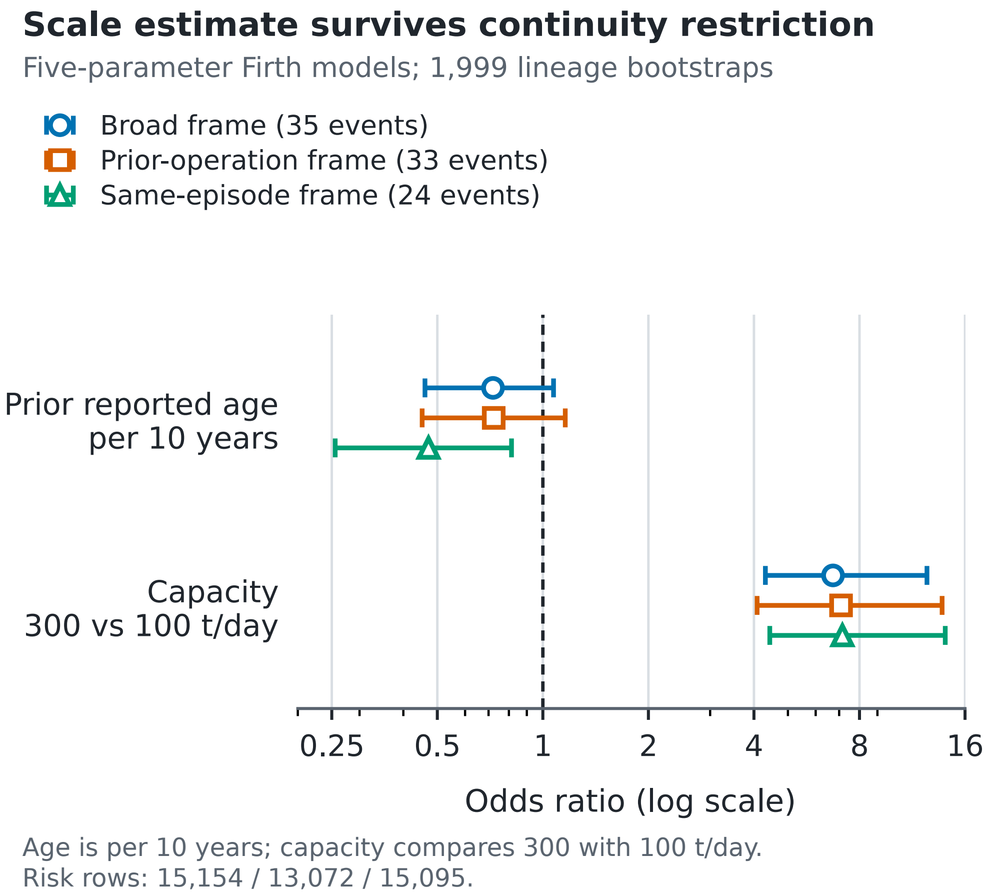

Figure 3: Five-parameter Firth estimates for first reported installed-capacity entry. Capacity is shown as the 300-versus-100 t/day contrast and age per ten years; intervals come from 1,999 whole-lineage bootstrap replications.

The supported RQ2 conclusion is narrow and stable across the reported sensitivities. First reported capacity entry is rare and selectively concentrated among larger waste-processing facilities. This concerns witnessed transitions among prior non-generators, not entry across the whole fleet. The administrative data do not establish a general independent age pattern, and scale can proxy for contracts, finances, catchments, technology, geography, or municipal capacity. Neither profile should be used as if the model had identified an equipment-project response.

4.3 RQ3: cohort differences align primarily with installed sizing

The engineering-valid generator frame has 6,511 observations from 493 stable lineages. Median gross intensity is 0.327 MWh/t, median generator design intensity is 14.0 kW per t/day, median electrical capacity factor is 0.607, and median waste-processing utilization is 0.609. These summaries describe generator-years that pass stated bounds, not the full fleet.

The RQ3 conclusion rests first on separate raw installed-kW and annual capacity-factor models. The exact component identity then explains how those independently reported quantities combine; it is not treated as independent evidence that sizing must dominate.

Reported start-year cohorts differ much more in installed generator size than in annual capacity factor. Relative to 2010 or later and conditional on observed controls, adjusted installed capacity is 79.1% lower before 1990, 58.6% lower in 1990–1999, and 23.5% lower in 2000–2009.

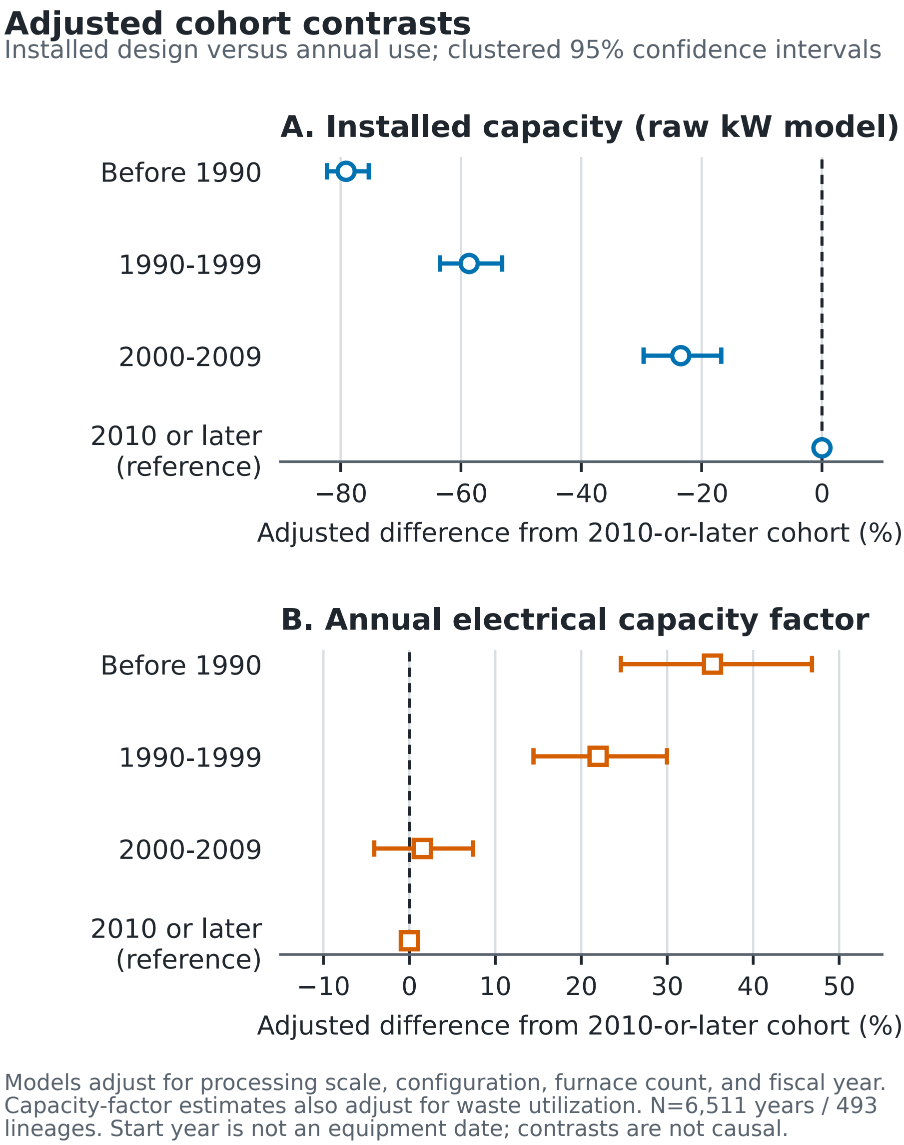

Figure 4: Adjusted cohort contrasts in installed electrical capacity and annual capacity factor. Points are percentage differences from 2010 or later; bars are lineage-clustered 95% confidence intervals.

The installed-kW elasticity with respect to processing t/day is 1.532 (95% CI 1.447–

1.617), and model R^2 is 0.786. Subtracting one gives the 0.532 design-intensity elasticity under identical controls; this is algebraic rather than independent replication.

The electrical-capacity-factor model tells a different story. Conditional on processing scale, waste utilization, technology, furnace count, and fiscal year, pre-1990 and 1990s capacity factors are 35.3% (95% CI 24.6–46.8%) and 22.0% (14.4–30.0%) higher. The 2000s estimate is 1.5% higher, with a CI from 4.1% lower to 7.4% higher. Waste utilization is positively associated with log electrical capacity factor (coefficient 1.695, 95% CI 1.448 to 1.942), while processing scale is negatively associated (-0.116, 95% CI -0.162 to -0.070). This model explains 33.9% of log capacity-factor variation. These coefficients describe annual use of installed kW. They do not show that utilization has an independent positive association with gross MWh/t after generator sizing is represented.

The direct gross-output model is consistent with the component structure. The elasticity of annual gross MWh with respect to throughput is 0.638 (95% CI 0.536 to 0.740), and the elasticity with respect to installed electrical capacity is 0.576 (95% CI 0.502 to 0.650), conditional on cohort, observed technology, furnace count, and year. The model R^2 is 0.914. It confirms that both waste loading and installed kW are central to annual output, but it is not a production function with exogenous inputs.

An accounting-consistency diagnostic compares gross-intensity specifications on the 5,806-row plausible-heating-value subset. In the legacy-style model without generator design intensity, the reported-age coefficient is -0.0349, the processing-capacity coefficient is 0.1001, and the waste-utilization coefficient is 0.6699; all have $p < 0.001$. After adding log generator design intensity, the corresponding coefficients become -0.0020 ($p = 0.2977$), -0.0092 ($p = 0.1991$), and -0.0995 ($p = 0.2038$). The sizing coefficient is 0.7532 ($p < 0.001$), and model R^2 rises from 0.4737 to 0.8131. This is not a causal decomposition, and the increase is partly expected because design intensity is algebraically related to gross MWh/t. The diagnostic shows that the former age, scale, and utilization interpretation is sensitive to whether installed sizing is represented; it does not independently prove that sizing causes the cohort pattern.

Adjacent-year rank persistence is highest for generator design intensity: 0.995 across 5,963 pairs from 470 lineages. Gross-intensity rank persistence is 0.961 and capacity-factor persistence is 0.873. The corresponding within-to-total variance ratios are 0.016 for log design intensity, 0.089 for log gross intensity, and 0.426 for log capacity factor. Sizing is therefore mostly a between-asset design attribute, while annual capacity factor contains much more within-asset movement.

The principal component conclusions remain stable across the two ten-year windows, lineage-equal weighting, and conservative or broad engineering bounds. For example, the processing-scale coefficient in the design-intensity model is 0.474 in FY2005–FY2014 and 0.577 in FY2015–FY2024; it is 0.520 under conservative bounds and 0.536 under broad bounds. Excluding every identity-uncertain lineage leaves 6,450 rows across 487 lineages and gives a nearly unchanged coefficient of 0.533. Within-asset-episode models also retain a positive association between utilization and electrical capacity factor. Those checks strengthen the component description but do not resolve simultaneous changes in throughput, maintenance, installed capacity, and output.

4.4 Exploratory extension outside the core research questions

This secondary analysis is outside RQ3 because the pathways do not verify physical project types and the followed groups are small. It is retained for hypothesis generation and professor discussion. The first-complete-year comparison asks where entrants sit in the same-year generator distribution, not what entry causes. Across all exact-year entrants with available engineering-valid outcomes at event time plus one, 44 lineages have mean gross-intensity rank 51.6%, generator-design rank 48.1%, and capacity-factor rank 56.3%. The pooled average is therefore near the middle of the contemporaneous generator distribution.

Pathways are heterogeneous. At event time plus one, 27 continuity-lineage entrants average 0.260 MWh/t and rank at 40.2% for gross intensity, 36.8% for generator design intensity, and 53.8% for electrical capacity factor. Eleven rebuild/replacement-like entrants average 0.442 MWh/t and rank at 72.5%, 66.1%, and 65.5%, respectively. Figure 5 displays these component ranks. Six forward-dated/placeholder observations are omitted from the plotted contrast because that category is too sparse and its administrative timing is difficult to interpret.

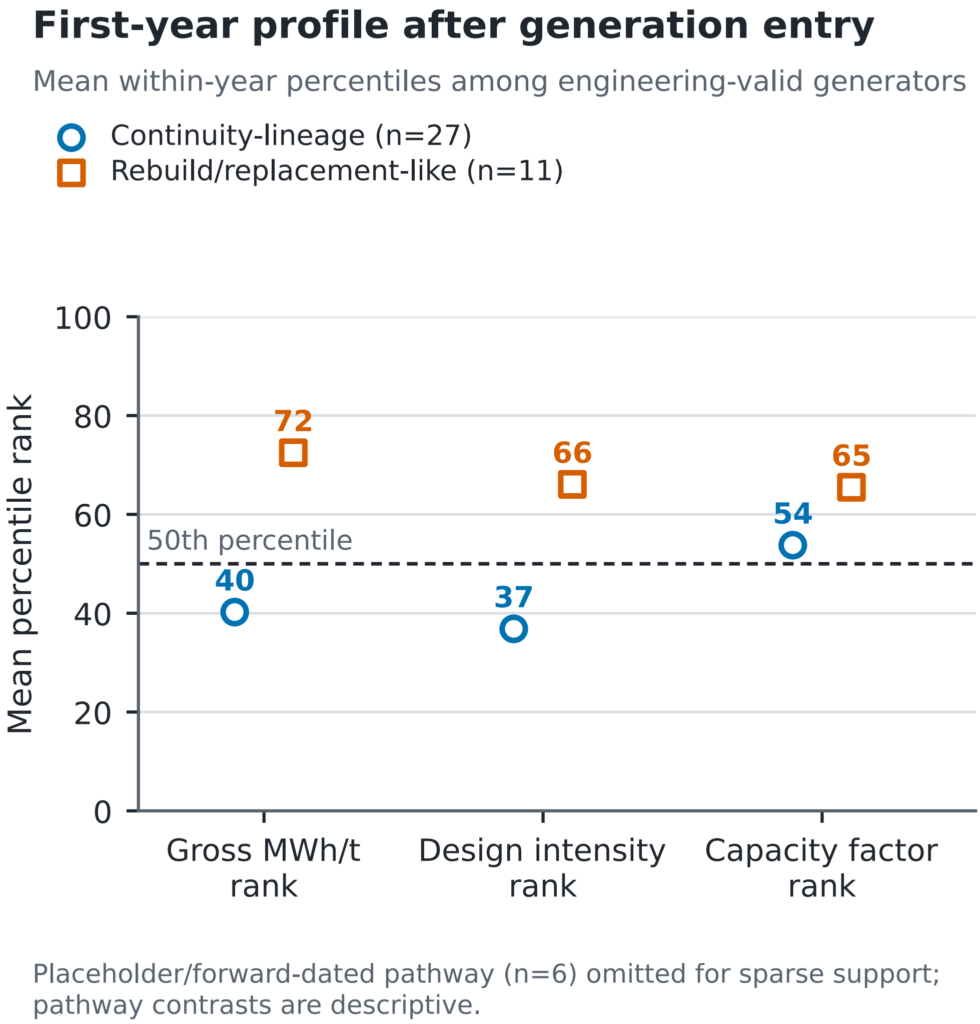

Figure 5: First-complete-year mean within-year percentile ranks for continuity-lineage and rebuild/replacement-like entrants. The contrast is descriptive; the sparse forward-dated/placeholder pathway is omitted.

The larger pathway difference aligns more closely with generator sizing than with capacity factor. That pattern is consistent with the broader component results, but it is not an estimated effect of pathway. Pathway assignment is based on reported resets, follow-up is selective, and there is no counterfactual match between otherwise equivalent projects.

5 Discussion

5.1 What the thesis changes in the fleet narrative

Installed-capacity participation is 41.1% among all records and 46.4% among positive-throughput records. Positive-output participation is 40.4% and 46.6% under the same two denominators. These same-state comparisons separate the effect of denominator choice from the small installed-capacity/output reporting difference. Positive-output facilities handle 80.1% of throughput, and the installed segment holds 70.5% of processing design capacity. Japan therefore combines incomplete diffusion across records with substantial volume coverage; non-generating records are not equal-sized unused opportunities.

Entry remains positively associated with scale under continuity, functional-form, reporting-state, geographic, and event-influence checks, but the model does not show that enlarging a facility would produce entry. Scale can proxy for unobserved municipal, technical, and financial conditions. Age is weaker evidence: individual estimates are negative, broad and identity-certain joint tests are non-significant, and the same-episode result lies at the conventional threshold with fewer events. This sensitivity is evidence against a simple age-only narrative, not evidence for one preferred continuity definition.

Gross MWh/t also changes meaning after decomposition. The separate raw-kW and capacity-factor models show that older cohorts have substantially smaller adjusted installed capacity but not lower annual capacity factors. The accounting diagnostic is consistent with those independent component results, but it is not their proof. Reported design and annual use point in different cohort directions.

5.2 Direct answers to the research questions

RQ1 is answered by matched state and denominator contrasts, not by one preferred percentage. In FY2024, installed capacity appears in 41.1% of all records and 46.4% of positive-throughput records; positive output appears in 40.4% and 46.6%, respectively. Positive-output facilities handle 80.1% of throughput, and installed-capacity facilities hold 70.5% of recorded processing design capacity. The difference means that generation is concentrated in larger or more heavily used parts of the observed fleet. It does not mean that 80.1% of waste becomes electricity, that the remaining throughput is recoverable, or that the covered facilities are environmentally optimal. RQ1 therefore changes the fleet narrative from "how many facilities generate?" to "which share of facilities, activity, and processing capacity is covered?"

RQ2 is answered by a scale association stable across the reported sensitivity analyses and a continuity-sensitive age association. The broad five-parameter Firth model gives a 300-versus-100 t/day odds ratio of 6.72, and the corresponding contrast remains close to seven under the prior-operation, same-episode, identity-certain, reclassification, and lineage-deletion checks. Alternative forms give 4.22–5.01, prefecture deletions 6.14–7.18, and the strict state frame 6.21. The evidence consequently supports processing scale as a stable profile of first reported entry in this administrative panel. Age is less stable across continuity definitions and should not support an age-only screening rule. Most importantly, the outcome is first reported positive installed capacity within an administrative lineage. It is not a verified retrofit, construction start, or causal response to changing capacity.

The absolute scale prevents overstatement: standardization gives approximately 2.53 versus 16.66 entries per 1,000 facility-years at 100 and 300 t/day. The association is large on the odds scale while the event remains uncommon.

RQ3 is answered by separating installed design from annual use. The adjusted cohort hierarchy is large for installed electrical capacity, whereas older cohorts do not show uniformly lower electrical capacity factors. Gross MWh/t therefore cannot be read as an independent operating-efficiency score. It is produced jointly by generator sizing, annual utilization of electrical capacity, and waste loading. The pathway comparison remains only a hypothesis-generating extension because it cannot answer RQ3.

Together, the answers provide one ordered diagnosis. RQ1 locates electricity recovery within the fleet; RQ2 describes which observed non-generator lineages first enter the installed-capacity state; and RQ3 explains the observable structure inside the generating segment. No result is asked to answer a question belonging to another frame. That separation is the central thesis contribution and the main safeguard against an aggregate or conditional result being interpreted causally.

5.3 Evidence-bound implications

Monitoring should report facility, throughput, and design-capacity coverage together. Screening can treat processing scale as a marker for further feasibility work, but not use an age-only rule. Existing generators should be compared conditional on sizing and design vintage before gross MWh/t is interpreted as an operating gap.

These are measurement and diagnostic implications, not a ranked list of projects. The study does not determine whether a municipality should build, replace, coordinate, maintain, or retire a facility. Such decisions require capital costs, waste-supply agreements, grid and internal-use conditions, maintenance history, emissions controls, heat demand, and alternatives higher in the waste hierarchy. The thesis's role is to prevent an aggregate statistic or misspecified ratio from deciding those questions implicitly.

5.4 Limitations and next evidence needed

The 20 workbooks are hash-identified and parser mappings are documented, but their original retrieval timestamps and publisher-side revision history are unavailable. Future acquisition should archive both.

Stable administrative lineages remain inferred. The resolver addresses exact duplicates, code breaks, row-order dependence, and known difficult links, but no physical-site registry confirms ownership, construction, or closure histories. Administrative absence is therefore not modeled as a physical outcome.

Entry is rare and partially left-censored. Firth estimation and lineage bootstrapping cannot overcome only 35 broad-frame, 33 prior-operation, and 24 same-episode events. Although all 1,999 requested bootstrap replications per frame converge, resampling cannot create missing project information. The parsimonious models omit financing, prices, contracts, catchments, maintenance, and detailed technology. Scale may proxy for these, so odds ratios are associations rather than effects of changing t/day.

Installed-capacity state is also a reporting construct. Forty-nine panel rows across six lineages report positive gross output despite a blank or zero installed-capacity field. None is the immediately prior row of a modeled event, and the two-prior-year sensitivity remains positive, but these checks do not prove that blanks represent physical absence. External equipment or commissioning records are needed to verify event meaning.

Generator fields also have strict boundaries. Gross generation is not net export; on-site use and useful heat are incomplete; reported capacity may differ from availability; and MWh/t is not the European Union R1 energy-efficiency indicator, a lifecycle measure, or an economic measure (Grosso et al., 2010). Heating-value coverage is insufficient for a common thermal measure. Bounds remove 149 rows but may exclude unusual valid cases or retain plausible-looking errors.

Reported start year is not an equipment date and can bundle original design, later equipment, reporting, waste, and municipality context. Annual throughput, capacity factor, maintenance, and output are jointly determined. Controls, fixed effects, and adjacent differences do not create exogenous variation; the equations remain accounting identities and conditional descriptions.

The 2010-or-later cohort enters only from FY2010 onward. Fiscal-year indicators create within-year comparisons where cohorts overlap, but cannot remove selective survival or unrecorded replacement.

The next evidence programme has three priorities. First, commissioning, procurement, or equipment records should verify the 35 modeled events and the administrative lineages around them. Second, net-export, internal-use, useful- heat, outage, cost, and waste-composition records should characterize project and operating performance. Third, only after policy exposure and credible comparison units are observed should a later study attempt causal evaluation of FIT, subsidy, retrofit, or replacement effects.

Finally, the pathway contrast is small and selected, and reported resets do not verify physical replacement. It remains a hypothesis-generating pointer toward the first and second evidence priorities, not a project-effect estimate.

6 Conclusion

Japan's fleet looks different at three margins. In FY2024, installed capacity appears in 41.1% of all records and 46.4% of positive-throughput records; positive output appears in 40.4% and 46.6%, respectively. Positive-output facilities handle 80.1% of throughput, and installed-capacity facilities represent 70.5% of processing design capacity. The broad 300-versus-100 t/day entry odds ratio is 6.72, corresponding to standardized annual predictions of 2.53 and 16.66 entries per 1,000 facility-years. The scale direction is stable across the reported sensitivity analyses, but age remains continuity-sensitive and neither association is causal. Older cohorts have markedly smaller adjusted installed kW but do not have lower annual capacity factors.

The thesis's contribution is therefore not a claim that every non-generator is an equal opportunity or that newer facilities simply perform better. It is a three-margin diagnostic: reconstruct lineages before transitions, separate counts from covered activity, and separate installed design from annual use. That architecture tells a professor or reviewer which claims require engineering, capital-history, or causal evidence.

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Contributor Roles Taxonomy (CRediT) Authorship Contribution Statement

Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry and e-Stat. Code, manifests, derived tables, figures, and the source workbooks used here are available at `https://github.com/Pann13223029/incineration-paper`. e-Stat permits reuse and modification with source citation under terms compatible with Creative Commons Attribution 4.0.

Declaration of Generative AI and AI-Assisted Technologies in the Thesis Preparation Process

During the preparation of this thesis, the author used OpenAI Codex and Anthropic Claude for language revision, thesis organization, and assistance with code development and review. The author executed the analyses, inspected the source data and generated outputs, independently checked the reported results, reviewed and edited all AI-assisted material, and takes full responsibility for the content. These tools were not used as authors and did not replace author judgment or accountability.

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